

ICAP

Intelligent Cognitive Alarm Platform

An AI-powered mobile application that builds healthy wake-up habits through cognitive challenges

Project Overview

The Intelligent Cognitive Alarm Platform (ICAP) is an AI-powered mobile application designed to help users develop healthy wake-up habits. Unlike traditional alarm apps, users must complete AI-generated cognitive challenges to dismiss alarms.

The platform also provides role-based dashboards for Users, Wellness Coaches, and Administrators to monitor performance, Habit Scores, and user progress.

KEY FEATURES



AI-Based Cognitive Challenges



Smart Alarm Scheduling



Automatic Habit Score Calculation



Role-Based Access Control



Performance Analytics Dashboard

My Contribution



My Role

Frontend Developer



Application Build

- Developed the complete React Native mobile application
- Designed responsive and user-friendly interfaces
- Implemented role-based navigation for User, Coach, and Administrator
- Integrated frontend with FastAPI REST APIs



Dashboards & Architecture

- Developed User Profile and Alarm Scheduling modules
- Developed Habit Score, Coach, and Administrator Dashboards
- Implemented analytics, monitoring, and dashboard visualizations
- Managed frontend architecture using reusable components

Technology Stack

Frontend



React Native

Expo



TypeScript

Type-safe development



Redux Toolkit

State management



React Navigation

Routing & navigation



Axios

API communication

Frontend Development

React Native Mobile Application — Developed Modules



Splash Screen



Login & Registration



Forgot Password



User Profile



Alarm Scheduling



Role-Based Navigation



Habit Score Dashboard



Coach Dashboard



Administrator Dashboard

Updated Frontend Application Enhancements



Habit Score Dashboard

- Replaced manual Habit Management with an AI-based Habit Score system
- Automatic score calculation based on challenge difficulty & completion time
- Weekly & monthly analytics with challenge history insights



Administrator Dashboard

- Coach assignment to users
- Complete user & coach management
- Habit Score and challenge performance monitoring
- Search, filter & sorting functionality



Coach Dashboard

- View assigned users & performance monitoring
- Habit Score analytics and challenge statistics
- Progress charts and individual user reports

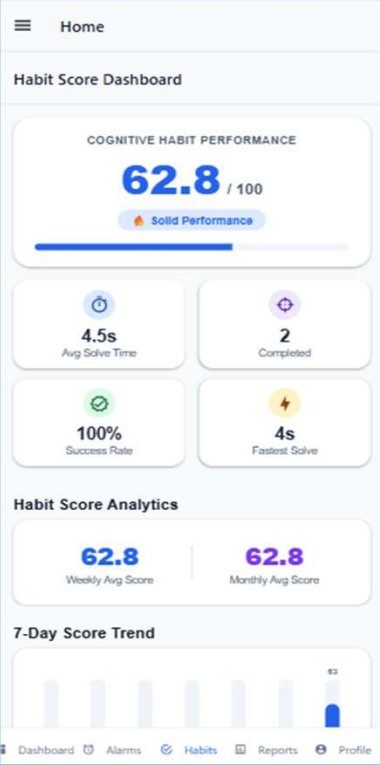


UI Enhancements

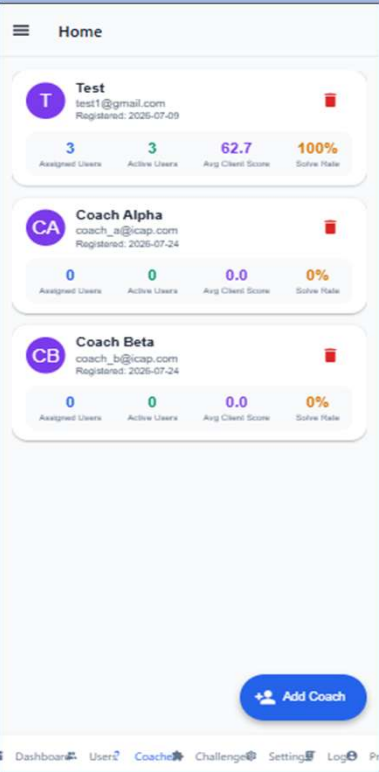
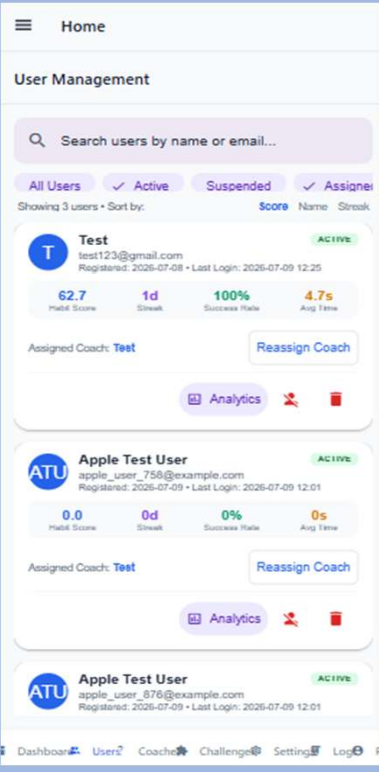
- Modern, interactive analytics charts
- Responsive card-based layout
- Improved navigation & real-time dashboard updates

Updated Frontend Application Snapshots

User



Admin



Coach



Future Work



AI Cognitive Challenge Engine

- Integrate the AI-powered Cognitive Challenge Engine
- Generate personalized cognitive challenges
- Dynamically adjust difficulty based on user performance



Application Testing

- Perform end-to-end functional testing
- Validate frontend and backend integration
- Conduct usability and performance testing
- Fix bugs and optimize performance



Deployment

- Deploy the FastAPI backend to a cloud platform
- Publish the React Native mobile application
- Configure production database and APIs
- Enable secure and scalable application access

Conclusion

- ✓ Successfully developed the frontend of the Intelligent Cognitive Alarm Platform using React Native
- ✓ Implemented secure role-based interfaces for Users, Coaches, and Administrators
- ✓ Built an AI-based Habit Score Dashboard with performance analytics
- ✓ Developed comprehensive monitoring dashboards for administrators and coaches
- ✓ Enhanced the application with real-time analytics, coach assignment, and user performance tracking
- ✓ Delivered a responsive, scalable, and user-friendly mobile application integrated with backend services